

Cramer's rule

- geometry behind solutions to lin sys equations
- Cramer's rule not the best, Gaussian elim better

$$\begin{aligned} 3x + 2y &= -4 \\ 1x + 2y &= -2 \end{aligned} \rightarrow \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -4 \\ -2 \end{bmatrix}$$

known matrix transforming unknown vector $\vec{x}$, & we know output is $\begin{bmatrix} -4 \\ -2 \end{bmatrix}$. which input vector $\begin{bmatrix} x \\ y \end{bmatrix}$ will land on $\begin{bmatrix} -4 \\ -2 \end{bmatrix}$?

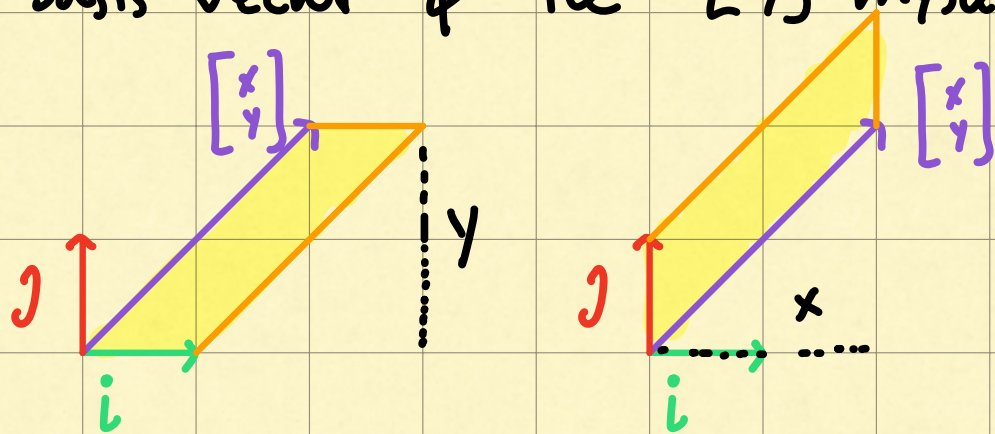
- answer u get depends on if transformation squishes all of space into a lower dimension (zero determinant) $\rightarrow$ either no sol or inf sol.

Say $\det(A) \neq 0$.

transformations which preserve dot products
are called orthonormal transformations,
like rotation matrices

is there a way to understand $\begin{bmatrix} x \\ y \end{bmatrix}$ unchanged
b/w the transformation?

think of parallelogram defined by each
basis vector & the $\begin{bmatrix} x \\ y \end{bmatrix}$ mystery input vector



$$A_{\square} = \text{base} \cdot \text{height}$$

$$= i_x \cdot y$$

$$= y$$

$$A_{\square} = j_y \cdot x$$

$$= x$$

- so now we think of coordinates as areas in 2D & volumes in 3D
- but now after applying transform, all areas get scaled by same amount, $\det(A)$
- Same $\Rightarrow$ as before now $\det(A) \cdot y$ & $\det(A) \cdot x$

• Now solve for coordinate y

$$y = \frac{\text{transformed Area}}{\det(A)} = \frac{\det\left(\begin{bmatrix} 2 & 4 \\ 0 & 2 \end{bmatrix}\right)}{\det\left(\begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}\right)}$$

2nd col is output vector

$$x = \frac{\text{transformed Area}}{\det(A)} = \frac{\det\left(\begin{bmatrix} 4 & -1 \\ 2 & 1 \end{bmatrix}\right)}{\det\left(\begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}\right)}$$

↙

1st col is output vector

Og. transform matrix

this is Cramer's rule